

CSE 461 : Introduction to Robotics

Lab Report

Lab No: 03

Lab Task: Introducing servo motor using along with Ultrasonic Sensor with Raspberry Pi

Group Information

Lab Section: 4

Group Number: 1

Group Members:

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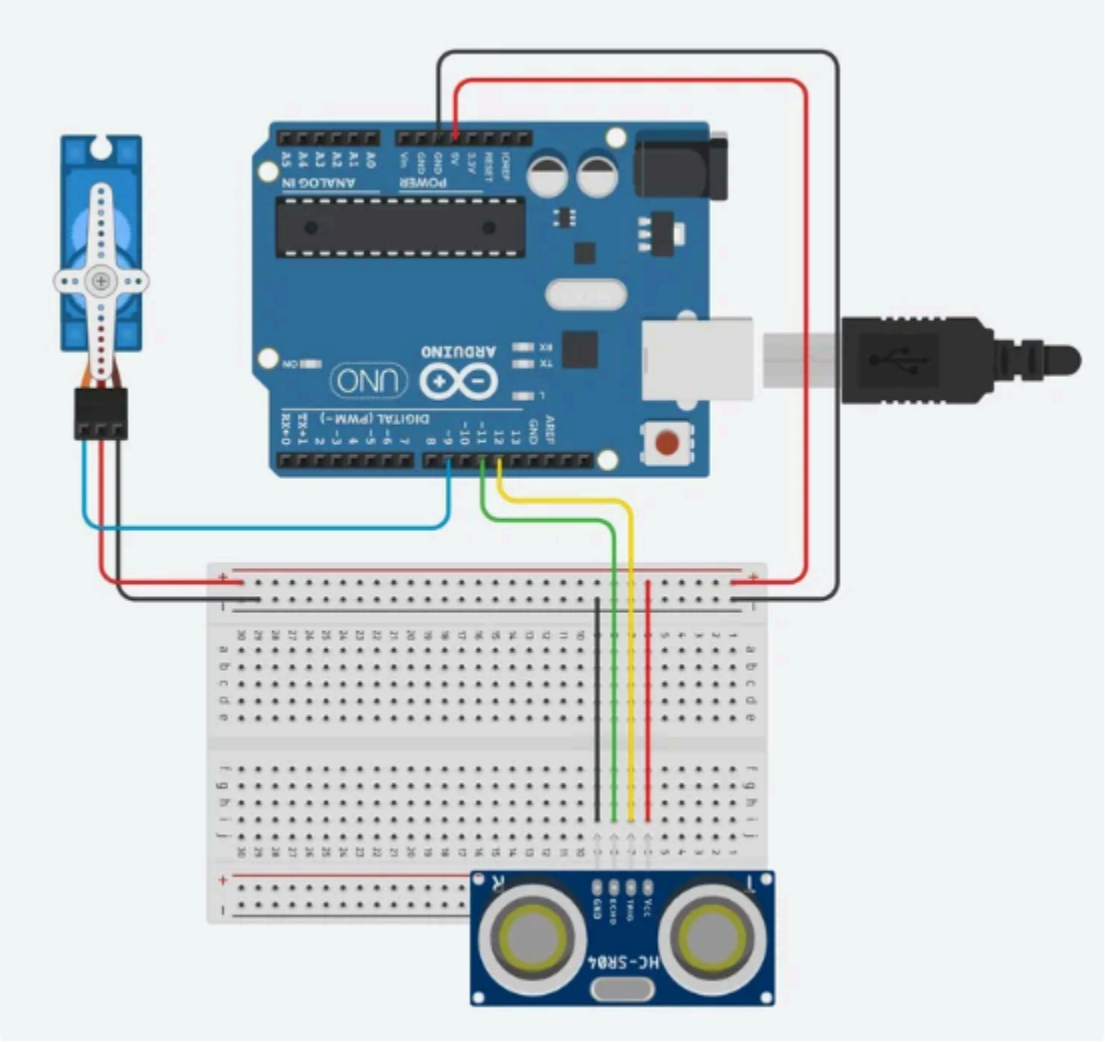
Task Description:

The main objective of this task is to know the functionality of the servo motor, rotating to a specific position, finding the proper duty cycle to achieve the desired angle for the motor and caliber with raspberry pi properly.

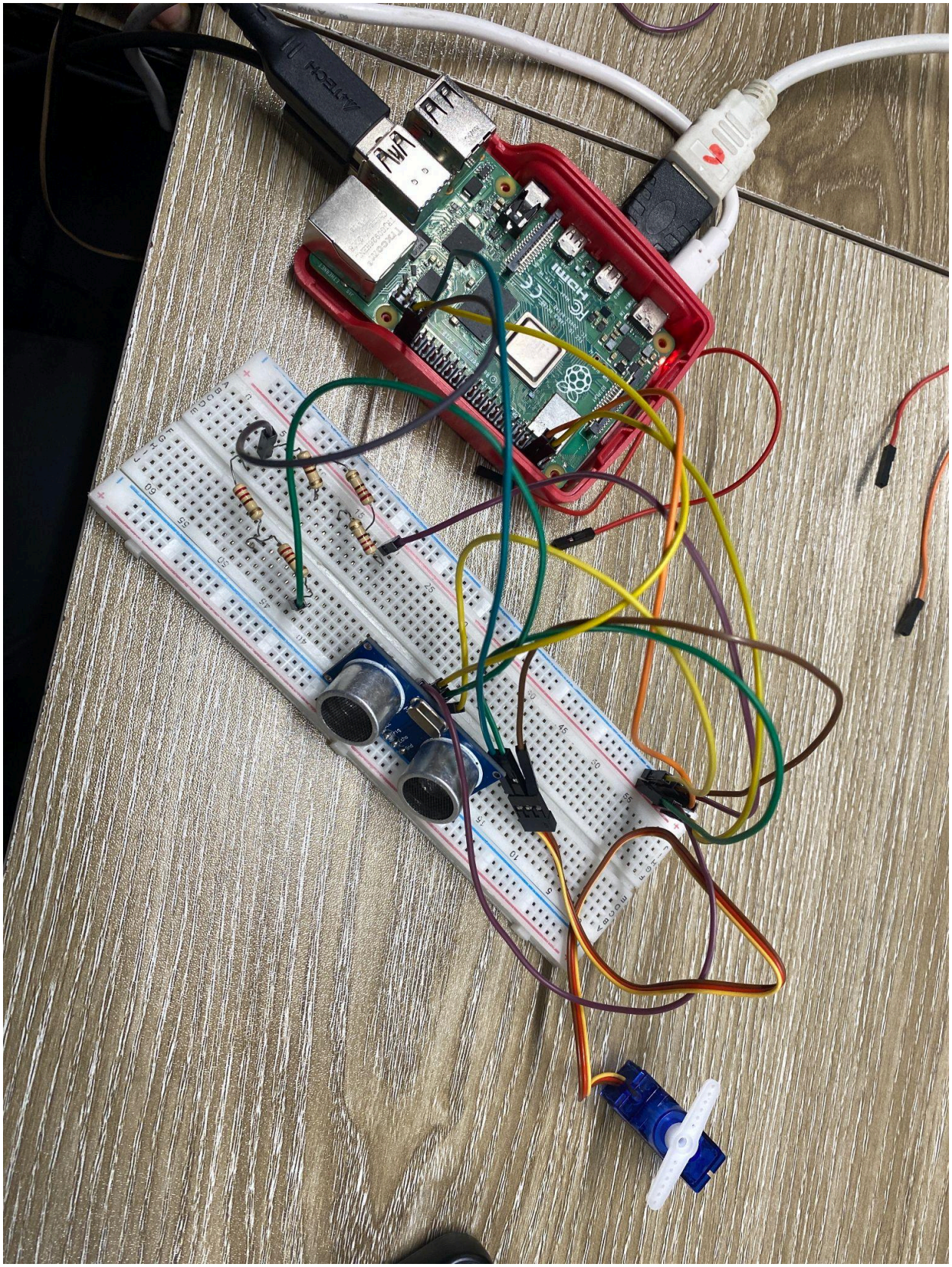
Components Used:

- Raspberry Pi 4
- MicroSD Card
- USB Cable
- Servo Motor
- Connecting wires (female to male)
- Monitor, Keyboard, Mouse(Optional)

Circuit Diagram:



Circuit Setup:



Code:

```
from gpiozero.pins.pigpio import PiGPIOFactory

from gpiozero import Device, Servo, AngularServo

from time import sleep

Device.pin_factory = PiGPIOFactory()

s = AngularServo(17,min_angle = 0, max_angle =
180,min_pulse_width=0.5/1000,max_pulse_width = 25/10000)

while True:

    s.angle=120# (120 degree to the left)

    sleep(1)

    #right

    s.angle=60 # 60 degree to the right

    sleep(1)
```

```
import RPi.GPIO as GPIO
from time import sleep


GPIO.setmode(GPIO.BCM)
GPIO.setup(17,GPIO.OUT)
pwm=GPIO.PWM(17,50)
pwm.start(7) #center(90 degrees)


while True:

    pwm.ChangeDutyCycle(5.3) # (left)

    sleep(1)

    pwm.ChangeDutyCycle(8.7) # right

    sleep(1)
```

Discussion / Q&A:

From this task, we gained practical experience working with Raspberry Pi GPIO pins and connecting them to servo motors. We learned how to interface a servo motor with the Raspberry Pi 4 using Python. One of the tasks was to calibrate the motor to associate it with 0 and 180 degrees. In order to make the servo motor rotate, we had to give power to the motor by connecting the Red wire to +5V DC power supply, Brown wire to the Ground and sending the PWM signals to the Orange colored wire. Since the “ON” time varies between 1ms to 2ms, so when the “ON” time is 1ms we checked the motor is at 0° means that the servo motor is the right position. Similarly, when the “ON” time was 1.5 ms the motor was at 90° or at center position and when it was 2ms it was at 180° or at fully left position. From this, we calculated the duty cycle for a specific degree. Overall, the project provided the team with a thorough understanding of GPIO pin functionality, servo motor integration, and the application of duty cycle calculations for motor control on the Raspberry Pi platform, equipping us with valuable insights and skills.